

# Kexin Chu

Ph.D. Researcher in ML Systems – University of Connecticut

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## SUMMARY

- Ph.D. student in Computer Science at UConn, focused on ML systems for LLM/MoE serving, KV-cache management, high-performance ANNS, and inference correctness.
- Research spans LLM serving efficiency, multimodal retrieval, and deterministic inference – with results published at and under review at top venues (ICDCS, SOSP).

## EDUCATION

|                                               |                     |
|-----------------------------------------------|---------------------|
| University of Connecticut, CT, USA (GPA:3.91) | Aug 2024 – Present  |
| Ph.D. in Computer Science and Engineering     |                     |
| Hefei University of Technology, China         | Aug 2017 – Jun 2020 |
| M.S. in Electrical Engineering                |                     |
| Hefei University of Technology, China         | Aug 2013 – Jun 2017 |
| B.S. in Electrical Engineering                |                     |

## PUBLICATIONS

- K. Chu, et al. *Dynamic OOD Graph ANNS for Multimodal RAG: Interference-Aware Continuous Repair*. Under review at SOSP 2026.
- K. Chu, et al. *MCaM: Efficient LLM Inference with Multi-tier KV Cache Management*. ICDCS 2025.
- K. Chu, et al. *SafeKV: System-Level Co-Design of Privacy Enforcement and KV-Cache Management for LLM Serving*. Under review at EuroSys.
- K. Chu, et al. *FlexMoE: Adaptive Expert Prefetching and Cache-Aware Routing for Fast MoE Inference*. Under review at ICCAD 2026.
- K. Chu, et al. *DynaExq: Runtime-Aware Expert Quantization for Efficient MoE Inference*. Under review at ICCAD 2026.

## RESEARCH PROJECTS (UCONN)

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| <b>Deterministic LLM Inference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mar 2026 – Present  |
| <ul style="list-style-type: none"><li>• Investigating sources of non-determinism in LLM inference across GPU parallelism, floating-point reductions, and runtime scheduling that cause identical inputs to produce different outputs across runs.</li><li>• Designing a system to enforce bit-exact reproducibility for LLM inference without sacrificing throughput, targeting production serving systems (vLLM/SGLang) and safety-critical applications.</li></ul>                                                                                                                                                                                                                                                                                   |                     |
| <b>Dynamic OOD Graph ANNS for Multimodal RAG</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Nov 2025 – Apr 2026 |
| <ul style="list-style-type: none"><li>• Identified that cross-modal (text-over-image/video) OOD queries require up to <math>9.2\times</math> more graph node visits than in-distribution queries, with ongoing dynamic updates continuously degrading the reachability structure.</li><li>• Designed an interference-aware continuous repair system with online repair-worthiness prediction, grouped local repair, and a latency-aware controller that bounds tail-latency impact during live serving.</li><li>• Stabilized recall at 93.7%–97.8% across three cross-modal datasets over 5-hour dynamic workloads; P99 latency spikes reduced by over <math>2\times</math> vs. periodic-repair baselines. <i>Under review at SOSP 2026.</i></li></ul> |                     |
| <b>SafeKV: Privacy-Aware KV-Cache Management for Shared LLM Serving</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Apr 2025 – Aug 2025 |
| <ul style="list-style-type: none"><li>• Designed a cache visibility control mechanism for multi-tenant KV reuse in shared LLM serving.</li><li>• Combined asynchronous privacy enforcement with metadata-based gating for cache promotion and reuse.</li><li>• Reduced timing-based KV side-channel attack success by 94% while improving throughput by <math>2.66\times</math> versus strict isolation. <i>Under review at EuroSys.</i></li></ul>                                                                                                                                                                                                                                                                                                     |                     |
| <b>DynaExq: Runtime-Aware Expert Quantization for MoE Inference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jan 2025 – Apr 2026 |
| <ul style="list-style-type: none"><li>• Identified that static expert quantization misallocates precision under a fixed HBM budget due to heavy-tailed, task-dependent activation; designed a runtime precision residency manager that promotes/demotes expert precision based on observed router traffic.</li><li>• Built non-blocking stream-parallel precision migration with pool-based memory management and admission control, keeping peak GPU usage within a hard 48 GB HBM cap.</li><li>• Recovered 79% of INT2-to-INT4 accuracy gap (73.58%→77.15% on Qwen3-MoE-80B) and achieved <math>2.73\times</math> throughput vs. offloading baselines. <i>Under review at ICCAD 2026.</i></li></ul>                                                  |                     |
| <b>FlexMoE: Adaptive Expert Prefetching for MoE Inference</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Aug 2024 – Apr 2026 |
| <ul style="list-style-type: none"><li>• Identified a U-shaped overlap–pollution tradeoff in fixed-horizon cross-layer prefetching that inflates end-to-end latency by up to <math>2.5\times</math> across batch sizes; reframed the prefetch horizon as a runtime control variable.</li><li>• Designed a closed-loop controller with EMA-smoothed stall/eviction signals, a CPU-side delta-adjusted expert forecaster, and a residency-aware two-tier cache with cache-aware token routing.</li></ul>                                                                                                                                                                                                                                                  |                     |

- Raised expert prediction accuracy by 21.8% and reduced cache-miss latency by  $13\times$  (geometric mean) across 3 MoE models on 3 accelerators. *Under review at ICCAD 2026.*

MCaM: Multi-Tier KV-Cache Reuse for LLM Inference

Nov 2023 – Aug 2024

- Built a multi-tier KV-cache system with prefix hashing, deduplication, and memory-aware GPU–CPU offloading.
- Reduced redundant prefill computation across requests with similar prefixes under bounded GPU memory.
- Achieved 69% lower prefill latency and  $3.3\times$  higher throughput in prototype evaluation. *Published at ICDCS 2025.*

INDUSTRY EXPERIENCE

Dcar, Inc. (ByteDance automotive platform)

Beijing, China

ML Systems Engineer Intern

Apr 2025 – Oct 2025

- Optimized the LLM serving system for Dcar’s AI Car Selection service, supporting 400K+ daily requests.
- Designed a resource-aware solution combining quantization and runtime offload/prefetch to address cross-node GPU inference inefficiency and oversized model memory footprints on a single  $8\times H20$  node.

Baidu, Inc. (China’s leading search engine company)

Beijing, China

Senior Software Engineer

Jul 2020 – Aug 2024

- Built and optimized Search Push & Recommendation (600M pushes/day), DeepQA Search (150M daily visits), and real-time streaming indexing pipelines; promoted twice (T3→T5) in 3 years.
- Reduced end-to-end data availability latency from 24 hours to 30 minutes and incremental indexing latency from 48 hours to 5 minutes via stream+batch pipeline redesigns.
- Reduced long-tail service latency by 63% through Go service decomposition and parallel retrieval.

TECHNICAL SKILLS

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Languages         | Python, Go, C++                                                                    |
| ML / Serving      | PyTorch, Transformers, vLLM, SGLang, CUDA, KV-cache reuse, deterministic inference |
| Systems / Backend | Linux, gRPC, FastAPI, Gin, Redis, MySQL, Docker, Git                               |
| Data / ANN        | Faiss, HNSW, DiskANN, IVF, OOD graph repair, Kafka, Hadoop, Spark                  |

AWARDS

Predoctoral Fellowship, University of Connecticut, 2025  
Baidu Pride Special Award, 2022  
Baidu Breakthrough Innovation Award, 2021–2022  
National Scholarship, Hefei University of Technology, 2018, 2019